📘 Project Summary

Project Title: Supermart Grocery-checkpointExploratory Data Analysis (EDA)

🎯 Objective

To analyze retail sales data to uncover trends, customer behavior, product performance, and profitability insights that can help guide business decisions.

📂 Dataset Overview

The dataset contains the following key columns:

- **Order ID** – Unique order identifier
- **Customer Name** – Name of the customer
- **Category / Sub Category** – Product classification
- **City / State / Region** – Geographical information
- **Order Date** – Date when the order was placed
- **Sales** – Revenue generated from the order
- **Discount** – Discount applied
- **Profit** – Profit made from the sale

🧹 1. Data Loading and Cleaning

- Loaded dataset using pandas
- Converted Order Date to datetime
- Removed duplicates and missing values
- Created derived columns: Year, Month, YearMonth
- Ensured numeric types for Sales, Discount, and Profit

📊 2. Descriptive Statistics

- Computed mean, median, mode, standard deviation for numeric fields
- Identified skewness and kurtosis in profit/sales distribution
- Displayed top 10 categories, sub-categories, cities, and states by frequency

📈 3. Time Series Analysis

- Resampled sales data monthly
- Calculated rolling averages
- Performed seasonal decomposition to identify sales trends

👥 4. Customer and Product Analysis

- Found top customers by sales and profit
- Analyzed sales by category, sub-category, city, and region
- Computed profit margins by category

🎨 5. Visualization

- **Bar Charts:** Sales by category, region, and sub-category
- **Scatter Plot:** Profit vs. Sales (with profit numbers shown)
- **Heatmap:** Correlation between Sales, Discount, and Profit
- **Line Plot:** Monthly sales trend

💡 6. Key Insights

- Certain categories have high sales but low profit — indicating possible over-discounting.
- Profit and sales are moderately correlated.
- Specific months show seasonal spikes in sales.
- A few top customers contribute significantly to total profit.

🧭 7. Recommendations

1. Review pricing and discount policies for low-margin categories.
2. Strengthen customer retention strategies for top customers.
3. Plan inventory and promotions for peak sales months.
4. Optimize logistics and marketing efforts in high-performing regions.